

SparkBridge cheat sheet

Every product on one page. List price, US dollars, per gateway, bought once and kept. Development and testing free with no time limit. Support optional at 20% of list per year. Nothing is charged per tag, per device or per connected client. This is the only price table in the course; every module links here.

Thesis in one line

Publish once at the site, add consumers at the Host, run the whole estate from one chair. Sites and consumers scale on separate lines. Nothing at a site changes when a consumer is added.

The core

Product	What it does	Runs where	Included with it	Price
SparkBridge Edge	Publishes site tags once on change over Sparkplug B, buffers to disk through a link drop, replays in order on source timestamps	Each site gateway	SparkLink (metered-link budget)	\$995
SparkBridge Host	Primary host. Collects every site into one namespace, builds tags from each site's birth, runs the Host modules	Host gateway, never at a site	SparkCalc, SparkID	\$1,495
SparkBridge Provider	Sparkplug-aware managed broker and policy layer: routing, per-site identities, deny-by-default topic permissions. Optional: any conformant broker the customer already runs works instead	Broker gateway or pair		\$1,995

Modules on Host

Each runs only on a SparkBridge Host gateway, on its own license.

Module	What it does	Status	Price
SparkCalc	Rates, totals and efficiencies calculated once on live data	Released	Included with Host
SparkID	Who can read and who can send commands, generated from intent; read never includes control	Released	Included with Host
Passage	Serves live plant data over OPC-UA to tools the customer already owns. Read only	Released	\$1,995
SparkInject	Production data landed in the warehouse exactly once, source timestamps kept. Per connector family: Snowflake first; Databricks and BigQuery follow	Released (Snowflake)	\$2,995 per family
SparkVault	Tag historian on the plant's own clock in an open database (Timescale); backfills trends after an outage	Released	\$3,995
SparkRecord	Tamper-evident record historian for structured events: Sparkplug records, MQTT JSON events, SparkFlow events, exactly once, SHA-256 audit chain. All destinations included	Early access 0.1.0	\$7,995
FleetOps	Every gateway in one view: versions, drift, certificates, backups	Released	\$4,995
Sentinel	Scored security and health assessment of the plant data network, with evidence and fix per finding. Observes only	Released	\$4,995
Sentinel compliance pack	Same findings as an auditor-ready binder. Add-on to Sentinel	Released	\$2,495
GitOps / Enterprise Governance	Configuration as code for the estate: signed bundles, review, rollback, audit trail	Pilot	\$9,995
SparkFlow	Every point streamed to the whole company past the tag-count ceiling (NATS backbone). Its own license	Released	\$995

Device drivers

Licensed per gateway. Each works with or without SparkBridge. Both Ignition lines (8.1 and 8.3).

Driver	Equipment	Price
SparkLogix	Allen-Bradley ControlLogix and CompactLogix over EtherNet/IP, live tag browse, no translator box	\$695
SparkSNMP	UPSs, switches, PDUs, servers over SNMP. One license covers both Ignition lines	\$695
IEC 60870-5-104	Utility and substation equipment, reads only	\$695
DNP3	DNP3 outstations with event data and source timestamps, reads only	\$695
TI 505 / CTI 2500	Older Siemens 505 and CTI 2500 racks, direct	\$695
SparkADS	Beckhoff ADS/AMS with live TwinCAT symbol browse	\$695
SparkS7	Siemens S7-1200/1500 symbolic addressing	\$695
Spark61850	IEC 61850 MMS client with report control blocks	\$695
SparkBACnet	BACnet/IP browse, read, write	\$695

Ignition enhancements

Licensed per gateway. Each works with or without SparkBridge.

Product	What it does	Status	Price
SparkInflux	InfluxDB history storage and history provider	Released	\$1,495
SparkValidate	Alarm rationalization (ISA-18.2) and validation document set (GAMP 5, Part 11, 62443). Rockwell first	Pilot	\$2,495 gateway module, \$995 Validate Desk CLI
SparkNotify	Alarm notification to Teams, Slack, WhatsApp, Telegram, Twilio SMS/voice, Discord, PagerDuty, acknowledgment tracked back	Released 0.2.0	\$695
SparkGantt	Perspective Gantt component, smooth at 500+ tasks	Released	\$695

Placement rule

Layer	What installs there
Site (EDGE)	Edge, SparkLink, any protocol driver the site needs
Broker	Provider, or the customer's conformant broker (HiveMQ, Mosquitto 2.1 tested)
Center (HOST)	Host, SparkCalc, SparkID, Passage, SparkInject, SparkVault, SparkRecord, FleetOps, Sentinel, GitOps, SparkFlow
Workstation	SparkID CLI, Sentinel scanner, Validate Desk CLI
Any gateway	Drivers, SparkInflux, SparkNotify, SparkGantt, SparkValidate module

Forty sites, three ways (list)

Shape	Required	Required total	Support / yr
Customer's own broker	39 Edge + 1 Host	\$40,300	\$8,060
Provider HA pair + history	39 Edge + 2 Provider + 2 Host + SparkVault	\$49,780	\$9,956
Complete operating architecture	Above, less Vault as required, plus Vault, FleetOps, Sentinel, GitOps	\$69,765	\$13,953

Site 41 is one more Edge, \$995. No number above moves when the estate gets busier.

Licensing terms, briefly

Unit	One license per production gateway, per product
Term	Perpetual
Development	Free, no time limit. Paid modules run on Ignition's two-hour module trial until a key file names them; Reset Trial restarts the clock
Support	Optional, 20% of list per year: updates and direct engineering support. Software keeps running if it lapses
Activation	One signed key file per gateway in the Ignition data directory, picked up within 30 seconds, no restart
Moving	Follows the role, not the hardware
Partners	Architecture Partners 20% off list with 90-day deal registration. First five Founding Partners 40% on their first production estate. No stacking

00 Orientation

What SparkBridge is, who buys it, and the frame every product fits into. Thirty minutes.

Learning objectives

- State the thesis in one sentence without notes.
- Name the three core roles and the four themes.
- Place any product in the line on the product map.

What SparkBridge is

SparkBridge is a line of modules for Inductive Automation's Ignition platform. GreenGuard USA built it for companies that run Ignition at many sites and want one system instead of forty. The customer already owns Ignition. We sell the modules that run on their gateways.

Two core products carry the line. **Edge** moves plant data reliably off each site. **Host** turns many sites into one system. **Provider** is the broker between them, sold when the customer does not already run a conformant MQTT broker. Everything else is an optional module that consumes the data Host holds, or a standalone driver or tool that works with or without SparkBridge.

The thesis

Publish once at the site, add consumers at the Host, and run the whole estate from one chair. Sites and consumers scale on separate lines. Nothing at a site changes when a consumer is added.

Every conversation comes back to this. When a prospect asks why the architecture matters, the answer is the thesis and then the number that proves it (module 05).

Six reasons the home page gives

1. Survive outages: Edge buffers to disk and replays in order.
2. Preserve source timestamps: a reading keeps the time the equipment gave it through buffer, replay, history and warehouse.
3. Scale horizontally: Host consumers share the stream with MQTT 5 shared subscriptions.
4. Eliminate polling: report by exception over Sparkplug B.
5. One namespace: Host builds the tag tree from each site's birth, no mapping file.

6. No per-tag pricing: per gateway, per product, bought once.

The four themes

Every module and every roadmap row lands in exactly one theme. Use them to organize a pitch and to answer "what else do you have" without reciting the catalog.

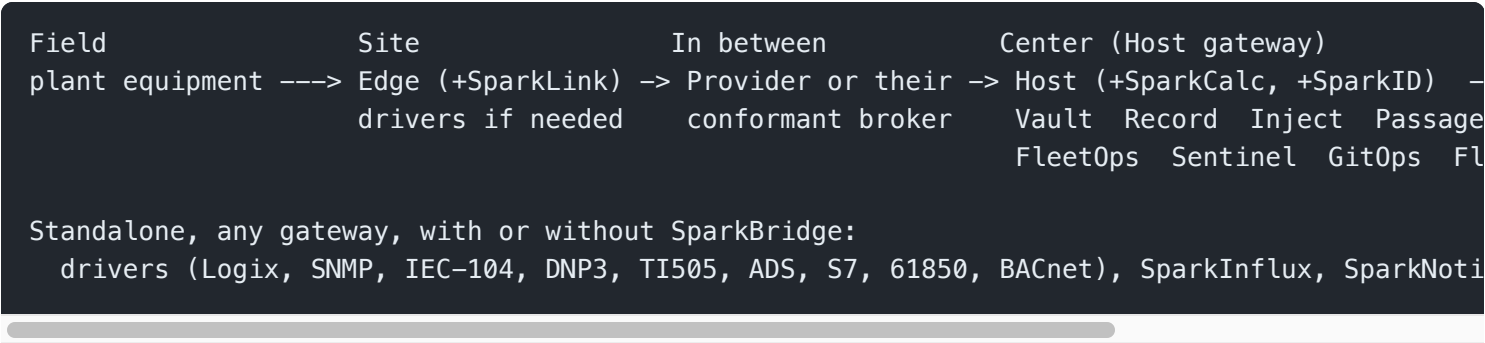
Theme	Promise	Products
Operate	Run a 40 to 200 site estate from one team without visiting sites	FleetOps, zero-touch enrollment, capacity intelligence, GitOps / Enterprise Governance
Trust	Prove where a number came from and that nobody changed it on the way	SparkID, Sentinel, data lineage (trace), data contracts, SparkValidate
Move Data	Land the same published data in every system the company runs, exactly or at-least once, with source timestamps intact	SparkVault, SparkRecord, SparkInject, Passage, SparkFlow, SparkNotify
Scale	Grow past what one Ignition server or one broker can hold, without a new integration project	Host pools and device sharding, Provider HA, federated estates

Who buys

Buyer	What they feel	Where to start
Controls or SCADA lead at a multi-site manufacturer	The hub gateway is the overloaded box; every new dashboard makes the sites work harder	Host + Edge, then SparkVault
Water, wastewater, utilities	Cellular and radio links drop; history has gaps; data bill climbs	Edge with SparkLink, then Vault
Oil and gas	Records must survive an audit; EFM and batch events	SparkRecord, SparkVault
Data centers and facilities	Network and building gear are not in the same picture as the process	SparkSNMP, SparkBACnet drivers
Validated industries	Weeks writing a validation package by hand	SparkValidate
System integrator with an Ignition practice	Wants a repeatable estate design and margin	Partner program (module 09)

Almost nobody buys everything on day one. Most plants buy one product for one problem, then add the next when it shows up.

The product map



Say this

- "Modules on the Ignition gateways you already license."
- "Each site publishes once. Every consumer subscribes at the center."
- "Per gateway, per product, bought once. Nothing per tag."
- "GreenGuard USA built it and measured it. The numbers are on the technology page."

Never say this

- "Replaces Ignition." It runs on Ignition.
- "Suite", "package", "bundle". There is no package. Each item has its own price.
- "Not included", "sold separately", "never bundled". Show the price line instead.
- Any hype word: seamless, robust, leverage, unlock, streamline.
- A reference customer we do not have. See module 11.

Check

1. Which two products are the core, and which is the optional third?
2. A prospect asks for "the security stuff." Which theme is that and which three products belong to it?
3. Where does SparkVault install: site, broker or Host?

► Answers

00a Ignition basics

For an SE who has sold automation but never installed Ignition. Sixty minutes plus a twenty minute lab. Inductive Automation (IA) publishes the platform; SparkBridge runs on it. Verify current edition names and prices against [inductiveautomation.com](https://www.inductiveautomation.com) before quoting anything about IA.

Learning objectives

- Explain gateway, Designer, client, tag, tag provider, UDT, driver, module in one sentence each.
- Describe how a module is installed and how the two-hour module trial works.
- Explain Sparkplug B to a controls engineer in under a minute.
- Avoid the "Edge" and "Host" naming collisions.

What Ignition is

Ignition is a SCADA, HMI and IIoT platform sold by Inductive Automation. One install is a **gateway**: a Java service on a server or industrial PC, configured through a web page (default port 8088, path /web/config). Engineers build screens and logic in the **Designer**, a desktop application that talks to the gateway. Operators use **Perspective** (browser and mobile) or **Vision** (Java desktop) clients. The gateway talks to PLCs through **drivers**, stores tag history in a SQL database, runs alarms and scripts, and connects to other gateways over the **Gateway Network**.

IA licenses per gateway, not per tag, client or device. That is why "unlimited tags, unlimited clients" is the phrase customers know. SparkBridge follows the same shape: per gateway, per product.

Editions

Edition	Purpose	Note for SparkBridge
Ignition	"Build any industrial application and deploy it anywhere." Per server, perpetual, unlimited tags, users and devices	Where Host and the Host modules run. Edge runs here at a site too
Ignition Edge	"Embed Ignition on edge-of-network devices." Per server, perpetual, unlimited tags and devices, at most 2 concurrent client views, 35 days internal data storage, no database connectivity, bulk discounts	IA's product named Edge, not ours. Ask "Ignition Edge edition, or the SparkBridge Edge module?" when the word comes up. Check Edge edition module support with engineering before promising SparkBridge Edge on it
Ignition Cloud Edition	"Add the power of the cloud to your Ignition system." Hosted on AWS or Azure marketplaces, pay as you go, usage-based	A Host can live here in principle; treat as a question for engineering
Maker	Free, non-commercial, home use	Do not sell against it or on it

IA support plans (annual, percent of retail at purchase): BasicCare 16%, TotalCare 20%, PriorityCare 24%. A major-version upgrade without a support plan is 65% of current retail. Redundancy needs both nodes licensed with matching modules. Source: [inductiveautomation.com/pricing/edition](https://www.inductiveautomation.com/pricing/edition), read 2026-09-06. Our 20% optional support sits next to TotalCare on the same sheet.

Vocabulary an SE must get right

Term	Meaning	Why an SE cares
Tag	A named value with quality and timestamp, in a folder tree	Sparkplug metrics map to tags; Host builds tags from births
Tag provider	A named tree of tags on a gateway; a remote tag provider exposes another gateway's tree over the Gateway Network	Remote providers are the fan-in load at the hub (module 00b)
UDT	User Defined Type: a template for a device with members and parameters	Edge admits a UDT whole or not at all; Sparkplug Templates carry it
Device connection / driver	The gateway-side connection to a PLC through a protocol driver module	Our nine drivers are this kind of module
OPC UA	Ignition's internal server exposes driver data; external clients can read it	Passage serves the Host namespace over OPC-UA to external tools
Tag history / historian	Tags marked for history are written to a SQL database through a history provider; 8.3 adds a Core Historian	SparkVault and SparkInflux are history providers
Alarm pipeline	Alarm events routed through notification pipelines	SparkNotify adds channels; SparkValidate rationalizes the alarms
Project	A set of screens, scripts and resources on a gateway	Not licensed; unlimited
Module	A signed .modl file installed at Config > Modules; adds drivers, components or services	Every Spark product is one
Gateway Network (GAN)	Gateway-to-gateway connection for remote providers, history and EAM	The traditional way to build an estate, and what we replace at scale
Store and forward	Gateway buffering of history writes when the database is unreachable	Different from Edge's disk buffer, which sits at the publish path
Redundancy	Master and backup gateway pair	Host active plus warm standby is our equivalent at the center

8.1 and 8.3

8.1 is the long-term support line most estates run. 8.3 changed how the gateway stores configuration (resources as files, a system.config API), added a secrets management system, event streams as project

resources, a Core Historian, gateway deployment modes, bundled JDBC drivers, and removed the serial modules into the core platform. On Siemens: symbolic S7 addressing that worked in the 8.3 early access was removed from the 8.3.0 release over last-minute licensing issues, and returned as the Siemens Enhanced driver, a separately purchased module not active by default. Module 04 has the detail and the SparkS7 pitch. Every Spark product ships on both lines, signed, and the 8.3 build is proven on a live 8.3 gateway before release. The customer's line decides which download they take; the price is the same.

How a module is installed and licensed

1. Upload the signed .modl at Config > Modules. The gateway shows the certificate and asks to trust it.
2. Without a key file a paid Spark module runs on Ignition's two-hour module trial. Reset Trial on the gateway restarts the clock. The data path pauses when the clock runs out. SparkLink, SparkCalc and SparkID never pause.
3. The customer drops one signed key file per gateway into the Ignition data directory. The module picks it up within 30 seconds, no restart, and the status page changes from Trial to Licensed.

Development and testing is therefore free with no time limit: reset the trial as often as needed. A license is bought when a gateway goes into production.

MQTT and Sparkplug B in sixty seconds

MQTT is a publish and subscribe protocol. Publishers send messages to topics on a **broker**; subscribers receive what they subscribed to. Sparkplug B is a specification on top of MQTT that fixes the topic namespace and payload so industrial systems interoperate. The pieces:

- **Edge node** publishes an **NBIRTH** with every metric's name, type and metadata, then **NDATA** with only what changed, by alias. Devices under it use **DBIRTH** and **DDATA**.
- The **primary host** is the one application that owns the namespace. Edge nodes watch its STATE topic, buffer while it is away and rebirth when it returns.
- Each metric carries its own timestamp, so source time survives the trip.
- Store and forward at the edge is the mechanism that makes an outage a delay instead of a loss.

IA's own MQTT modules (Cirrus Link MQTT Transmission, Engine and Distributor) are the reference customers compare us to. SparkBridge Edge is a Sparkplug edge node, Host is a primary host, Provider is a Sparkplug-aware broker. All three interoperate with conformant third parties by specification, and the site says honestly which pairings have run on the bench.

Naming collisions

- **Edge.** IA sells "Ignition Edge" (an edition). We sell "SparkBridge Edge" (a module). Always say the full name once per conversation.
- **Host.** In Sparkplug the "primary host application" is a role. Our product named Host is that role plus the modules. IA's Cirrus "MQTT Engine" plays the role on their side.
- **Provider.** In Ignition a "tag provider" or "history provider" is a data source. Our Provider is a broker. Say "SparkBridge Provider, the broker" the first time.

Lab, twenty minutes

Use the GreenGuard lab gateway (Ignition 8.1.54 at localhost:8088 on the Mac, or the 8.3 gateway on gg-wsl). Read only; change nothing.

1. Open Config > Modules. Find a SparkBridge module, note its version and the certificate it was signed with.
2. Open the module's status page (Status > SparkBridge or the Config > SparkBridge tab). Note whether it says Trial or Licensed and when the trial resets.
3. Open Designer, browse the default tag provider, find a UDT instance and open its definition.
4. On the Host page, open a connected edge node and read its last birth: metric count, timestamp, device list.
5. Write down the one sentence you would say to a prospect about each of the four things you saw.

Say this

- "Modules on the Ignition gateways you already license."
- "Same per-gateway model as Ignition itself."
- "Free to develop and test, for as long as you need."

Never say this

- "Replaces Ignition" or "SCADA in the cloud."
- "Works on any edition." Confirm the edition first.
- Anything about IA pricing from memory. Point to IA's site.

Check

1. What does the gateway do when a paid module's two-hour trial runs out, and what fixes it without a restart?
2. What is the difference between Ignition store and forward and the SparkBridge Edge buffer?
3. Which Sparkplug message carries metric names, and which carries only changes?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

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00b Ignition at scale

How Ignition estates are actually built at 1, 5, 20 and 40 sites, and where each shape starts to hurt. This module is the bridge between knowing Ignition and knowing why SparkBridge exists. Forty-five minutes.

Learning objectives

- Draw the four deployment shapes and name the boxes someone has to license, patch and watch.
- Say what breaks in each shape at 5, 20 and 40 sites.
- Map a described estate to a shape and to the SparkBridge required set.

The four shapes

1. Single gateway

PLCs ----> one gateway (drivers, tags, history, screens, alarms)

One plant, one server. Nothing to fix. Most customers start here and it is fine at one site.

2. Hub with remote tag providers over the Gateway Network

```
site gateway 1 --GAN--+
site gateway 2 --GAN--+--> hub gateway (remote tag providers, central screens, history)
site gateway N --GAN--+
```

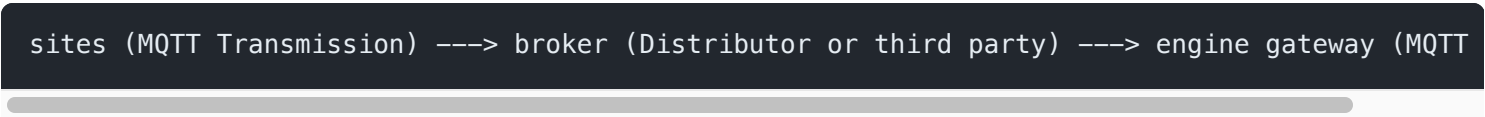
Each site keeps its gateway. The hub subscribes to each site's tag tree as a remote provider. Every remote provider is a live subscription load on both ends. Every consumer at the hub (screens, history, a report) is more subscription load back onto the sites.

3. Front-end and back-end split with redundancy

```
sites --GAN--> backend A (history)    backend B (applications)    backend C (integrations)
                  |                    |                    |
                redundant pair        redundant pair        redundant pair
                  \_____ front-end gateways (clients) ____/
```

When the hub saturates, the estate is split into backends by application family, each holding its own copy of the estate, each redundant. Site count multiplies backend load; consumer count multiplies site load. History is written per backend and reconciled by hand.

4. Module-based MQTT



Sites publish over Sparkplug and buffer through outages. Polling and most fan-in are gone. What is left to custom engineering is everything after the transport: consumer scale-out (a second consumer usually means a second engine), provisioning (broker URL, credentials and certificates typed at each site), configuration drift, fleet health, security policy written as ACL files by hand.

What breaks, by site count

Sites	Shape 2 or 3 (traditional)	Shape 4 (module MQTT)	SparkBridge
5	One hub, remote providers. Fine	Fine	5 Edge, 1 Provider or their broker, 1 Host. Also fine
20	Hub CPU and memory climb with every remote provider; history moves to a second backend	Transport fine; a second consumer needs a second engine; site setup is manual	Same three roles. Host memory grows about 300 bytes per wire metric per consumer
40	The hub is the overloaded box. A second hub splits the estate and every consumer now needs both	Fleet operations are the problem: which sites are offline, buffering, drifted, expiring	Provider HA pair, Host active plus warm standby (takeover measured at 244 ms across 25 nodes). One namespace
50 and beyond	A backend per consumer family; integration projects	Custom tooling for everything after the broker	Add Host consumers on the shared stream; sites never change

The real case on the use-cases page: one central server carrying 40 sites at 50 to 75% CPU with 14 GB of memory, unable to ride out a network outage, getting worse with every dashboard. Per-site scale in the migration plan that produced the reference architecture: Irving 8,700 published leaves, Mesa 22,400, Dickinson 8,500, fleet average about 13,000 per site.

The six problems that arrive with site count

Problem	Traditional	Module MQTT	SparkBridge
Gateway Network fan-in	One hub carries every site connection and saturates first	Removed: sites publish to a broker	Removed: sites publish to Provider; Host consumers share the stream
Backend sprawl	A backend per application family	One engine; a second consumer usually means a second engine	One Host pool; consumers are modules on Host or subscribers of the same stream
History duplication	Site history plus central history, reconciled by hand	Depends where history is enabled	History is one consumer. SparkVault writes once, centrally, on source timestamps
WAN outages	History during the gap depends on per-site store-and-forward configuration	Store and forward at the site	Disk buffer at 2,697 messages/s durable; 400,000 of 400,000 replayed in order with the historical flag so trends backfill
Timestamps	Arrival time unless every hop is configured to keep source time	Source timestamps by Sparkplug	Source timestamps through buffer, replay, history and warehouse
Provisioning and fleet	Manual gateway build, GAN approval, provider setup per site; status pages one at a time	Credentials typed at each site; broker client list	One-time enrollment code over a signed channel; FleetOps versions, drift, certificates, backups in one view

Where a module-MQTT cell reads the same as ours, that is the Sparkplug standard doing its job for both. The difference is what remains for the customer's team to build afterwards. Say that out loud; it is on the compare page and it is honest.

Two per-gateway bills on one sheet

IA charges per gateway, unlimited tags and clients. SparkBridge charges per gateway, per product, nothing per tag. An SE can put both on one sheet for a 40-site estate: 40 Ignition site gateways the customer already owns or will buy from IA, plus 39 SparkBridge Edge, 1 or 2 Host, and Provider if they have no broker. Neither line moves when tag count grows. That single fact removes the objection "what happens to the price when we add tags."

Discovery questions that reveal the shape

1. How many gateways run today, and which are redundant pairs?
2. Where is history written, and how many databases hold a copy of the same tag?
3. How do sites reach the center: Gateway Network, MQTT, VPN, nothing yet?
4. What happens to history and alarms during a WAN outage at a site?
5. Who builds a new site gateway, and how long does it take from box to data at the center?
6. How many consumers of site data exist: screens, historians, reports, warehouse feeds, analytics?
7. What is the CPU and memory on the busiest central box right now?

Answers to 1, 3 and 6 place the estate in a shape. Answers to 4 and 7 give the number that opens the SparkBridge conversation.

Say this

- "Site count multiplies hub load. Consumer count multiplies site load. We cut both lines."
- "Sparkplug removes polling for everyone. What we add is what comes after the transport."

Never say this

- "Ignition doesn't scale." It does; specific shapes hurt at specific sizes.
- "MQTT modules are wrong." They are the right transport, and we interoperate with them.

Check

1. A prospect runs 18 sites on a hub with remote tag providers and a separate history backend. Which shape, and what is the first symptom they will describe?
2. Same prospect. Write the SparkBridge required set and its list total.
3. A prospect already runs MQTT Transmission at every site into HiveMQ. What is left for them to build, and which two of our products address it first?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

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01 Edge, Provider, Host

The three roles. What each does, what runs where, and the facts an SE can state without checking. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Explain each role in two sentences.
- Answer "do we need Provider" correctly.
- State the five measured numbers that belong to the core.

SparkBridge Edge, \$995 per site gateway

Edge is the module installed at each site. It watches a folder of tags the customer chooses, sends each value once when it changes with the timestamp the equipment gave it, and keeps recording to a disk buffer when the link drops. On reconnect the backlog arrives in order and lands on the time it actually happened.

Fact	Detail
Where it runs	The site gateway, nowhere else. Standalone; nothing else installs at the site
Tag structure	Read from the site's own folder including UDTs, announced to Host in the birth. No mapping file. Folders at a chosen depth become devices
Tag budget	250,000 per gateway. A UDT is admitted whole or skipped whole. Subscriptions install in 10,000-path chunks
Edge nodes	Up to eight per gateway, each with its own connection, tag set and buffer
Buffer	50 MB default per edge node; size cellular and satellite sites at 500 MB. Measured 2,697 messages/s durable
Write-back	Off by default; allowed only for listed devices and points
Broker	Provider, or any conformant MQTT broker (HiveMQ, Mosquitto 2.1 tested), TLS and per-site credentials
Enrollment	Zero-touch: one-time code over a signed TLS channel; credentials written owner-only, never typed at the site. 2.9.0 adds CSR/CA and mTLS
Ignition	8.1.19 or newer, and the 8.3 line
Included	SparkLink: measures what a metered site spends, attributes it to the tag, eases off gradually when spend runs ahead, never touches points marked safety-relevant

SparkLink story worth telling: in a real three-day capture one vibration tag with no change threshold set was taking 50.7% of a site's monthly data budget, 47 times the next largest consumer.

SparkBridge Provider, \$1,995 per gateway

Provider is SparkBridge's Sparkplug-aware broker and policy layer: routing, per-site identities, deny-by-default topic permissions, no client count. It runs as an embedded Moquette or as a HiveMQ CE companion service. Two Provider nodes make the HA pair in the reference architecture (4 cores, 8 GB each).

Do we need Provider?

No, if the customer already operates a conformant MQTT broker they are happy to keep. Edge and Host connect through it. Yes, if they have no broker, want SparkID's generated permissions enforced without writing ACL files, or want us to own the HA pair. Say it positively: "Provider, or the broker you already run."

SparkBridge Host, \$1,495 per Host gateway

Host is the primary host. It collects what every Edge reports, learns each site's tag structure from its birth and projects one live tag tree that mirrors the plant. Add a device at a site and it appears at Host. When a site drops, Host marks that site's points bad rather than holding the last number, and asks for a fresh birth on return. A sequence gap triggers the same check.

Fact	Detail
Where it runs	Host gateway, never at a site. Single Host, or active plus warm standby, or a pool
Speaks	Sparkplug B and 3.0 over MQTT 3.1.1 and MQTT 5, as the primary host
Consumers	MQTT 5 shared subscriptions: four Host consumers deliver about 20x one. BIRTH, DEATH and STATE are subscribed non-shared so alias tables stay correct
Memory	About 300 bytes per wire metric per consumer; 500,000 wire metrics in under 4 GiB
CPU	Projector shards equal core count. Six cores sustained 339,527 updates/s end to end on the Linux reference rig
Takeover	Standby promotes in 244 ms across 25 nodes; edges never rebirth
Capacity route	Host reports its own ceiling from benchmark evidence and the shard count it runs (/capacity), not a fixed number
Fleet route	/fleet shows buffering and headroom per edge; Prometheus metrics
Recommended box	8 cores, 16 GB; 32 GB if SparkVault shares it
Included	SparkCalc and SparkID
Ignition	8.1.19 or newer, and the 8.3 line

Host alone, no modules added, replaces the work of stitching forty sites together by hand. The modules are how the customer opens that data, puts numbers on it, secures it, runs it and keeps it. Each needs a Host gateway underneath it.

Cross-cutting facts, since 2.7.0

- Tiered route authorization: every status and control route on every Spark product uses the same read, write and admin token policy. Loopback trust is off unless turned on. A forwarded header cannot impersonate the local host.
- One signed key file per gateway, checked every thirty seconds, applied with no restart. An invalid or tampered key fails closed.
- Data lineage: six trace hops shipped in 2.8.0; wire hops arrive in 3.0.0. Data contracts shipped in 2.8.0.

The five numbers

Number	Claim	Source page
2,697 messages/s	Durable disk buffering at Edge	technology, compare
400,000 of 400,000	Replayed in order after an outage with the historical flag	compare, benchmarks
244 ms	Host standby takeover across 25 nodes	compare, architecture
500,000 under 4 GiB	Wire metrics held in Host memory	compare, technology
About 20x	Four shared-subscription consumers versus one	compare, benchmarks

Cite the page. Every figure was measured on named hardware and the reproduction steps are in the reference architecture document.

Say this

- "Edge at each site. Host once. Provider, or the broker you already run, in between."
- "A dropped link is routine. Edge was built around that fact."
- "Host marks a site bad instead of stale. Nothing quietly drifts."

Never say this

- "Unlimited tags at Edge." The budget is 250,000 per gateway.
- "Provider is required." It is not, if they have a conformant broker.
- "Mixed-vendor broker pairing is certified." It is by specification; say which pairings ran on the bench (HiveMQ, Mosquitto 2.1).

Check

1. A site has 300,000 candidate tags. What does Edge do?
2. The customer runs HiveMQ Enterprise. Which of the three core products do they buy?
3. What happens at Host when a site's WAN drops, and what happens when it returns?

► [Answers](#)

[Previous: 00b Ignition at scale](#) ·

02 Data consumers

The Move Data theme: the modules that take what Host holds and land it somewhere useful. Each runs on the Host gateway on its own license. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Match each consumer to the customer sentence that triggers it.
- Explain SparkVault versus SparkRecord in one breath.
- State each module's status label exactly as the site does.

SparkVault, \$3,995, released

Trigger: "A trend runs flat through an outage and then jumps. Nobody can defend the report built on it."

SparkVault is the family's tag historian: time series, trend readback and replay, in an open database (TimescaleDB) served back as an Ignition history provider. History is one consumer of the backbone; adding it changes nothing at the sites. Every reading lands on the time it actually happened, so the backlog fills in where it belongs after an outage. Any past day can be replayed to test a dashboard or walk through an incident.

Proof: a gateway disconnected for four hours and eleven minutes while production continued; zero values lost, every reading on its original time. Host-side only; nothing installs at the edge, which is why there is no "site package."

SparkRecord, \$7,995, early access 0.1.0

Trigger: "Our batch and event records have to survive a rerun, a migration or an auditor."

SparkRecord is the structured record lane: Sparkplug B records (Cirrus Transmission Record objects, DataSet and Template metrics), JSON events on arbitrary MQTT topics, and SparkFlow events over NATS (the NATS lane ships in 0.2.0). Schema is discovered from the payload and migrated additively only. Delivery is exactly once: source and receipt timestamps kept separately, producer identity on every row, deduplication by content hash, ledger written in the same transaction. Every record id is a SHA-256 over canonical JSON chained to the one before; the module never issues UPDATE or DELETE. Destinations: PostgreSQL, SQL Server, TimescaleDB, Snowflake, S3-compatible object storage, all included.

Vault or Record

SparkVault stores continuous telemetry: timestamp, metric, value, quality. SparkRecord lands structured events: a batch, a transaction, an alarm record, with a tamper-evident chain. Most estates need the first; regulated ones usually need both.

Say "early access 0.1.0" every time. Do not promise the NATS lane or a date beyond what the roadmap page says.

SparkInject, \$2,995 per connector family, released for Snowflake

Trigger: "Analysts reconcile duplicate rows in the warehouse by hand and never trust the totals." Or: "We pay a per-row subscription to see our own production data."

SparkInject streams plant data into the customer's own Snowflake account, exactly once, with source timestamps kept and an offset token per row. The customer owns the account, the storage and the bill; we are not in the middle. A restart or a dropped connection cannot cause a gap or a duplicate. Proof: 8,375 readings queued through a four-hour outage and landed with nothing missing and nothing counted twice.

Connector families: Snowflake first. Databricks and BigQuery follow. Kafka, ADX and S3 sinks were live-proven in the 2.7.0 bundle; quote them only after checking the current products page for what is listed as available. One family per license.

Passage, \$1,995, released

Trigger: "We have a historian or SCADA package that will never be replaced and it still needs the same live data."

Passage serves the Host namespace over OPC-UA so the tools the customer already owns connect the way they already do. No custom connector per system. Read only: it cannot write or command, which is a selling point to the security team.

SparkCalc, included with Host, released

Trigger: "The same efficiency number is hand-built in a PLC, a dashboard and a spreadsheet, and gives three different answers."

Rates, totals and efficiencies calculated once, in one place, on live data, then used everywhere. Runs at a site or centrally.

SparkFlow, \$995, released, its own license

Trigger: "Screens and analytics for the whole company need every point, live, and the tag system is the ceiling."

SparkFlow streams every point Host decodes to a NATS backbone, past what the tag system carries, lossless and replayable. Built for the ultra-large estate. Priced on its own, on any shape.

SparkNotify, \$695, released 0.2.0, any gateway

Listed here because it moves alarms rather than data: one alarm to Teams, Slack, WhatsApp, Telegram, Twilio SMS or voice, Discord and PagerDuty from one configuration, acknowledgment tracked back into Ignition. Secrets encrypted at rest on 8.3. Works with or without SparkBridge. Email channel is removed in 3.0.0; do not sell it.

The consumer add story

Each of these is a subscription at Host or a module on it. No change at the forty sites. When a prospect asks about a future analytics or AI project: subscribe to the shared stream, read SparkVault, or take the Snowflake copy. A second corporate historian: Passage already serves OPC-UA, add a client. That sentence is the thesis applied.

Say this

- "History is a consumer, not another integration project."
- "Exactly once, on the plant's own clock, in a database you own."
- "SparkRecord is early access. Here is what 0.1.0 does today."

Never say this

- "SparkVault replaces the site historian." It is central; a site may keep local history for local screens.
- "Inject supports every warehouse." Per family; Snowflake today.
- "Passage can write." It cannot.

Check

1. A pharma customer wants batch records with an audit trail and also long-term trends. What do they buy?

2. A customer uses OSIsoft PI and wants it fed from the same data. Which module, and what is the security answer?
3. What does a customer buy to land data in both Snowflake and Databricks?

► **Answers**

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 01 Edge, Provider, Host](#) ·

03 Operate and trust

The Operate and Trust themes: running the estate from one chair, and proving where a number came from. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Explain FleetOps, Sentinel, SparkID, GitOps and SparkValidate by the question each answers.
- State which are included, which are pilot, and what is roadmap.

FleetOps, \$4,995, released, Host

Question it answers: "Forty gateways in the field and nobody knows which version, which drifted, or which certificate expires next month."

One view of every gateway: versions, drift, certificates, backups. A fix is made once and applied everywhere instead of forty visits. Worth it from roughly ten sites up. True fleet management (config templates, drift remediation) is a 3.1 program; say "drift is shown per gateway" and stop there. 2.9.0 registers enrolled sites into FleetOps automatically.

Sentinel, \$4,995, released, Host; compliance pack \$2,495

Question it answers: "Nobody can say what the plant network is exposing, and an auditor is asking for evidence."

A scored security and health assessment of the plant data network, ranked by risk, with the evidence behind each finding and the fix for it. It only observes and changes nothing. The compliance pack turns the same findings into an auditor-ready binder. The scanner also runs from a workstation.

SparkID, included with Host, released

Question it answers: "One compromised site could write another site's data, or send a command to equipment it has no business touching."

SparkID generates deny-by-default topic permissions from intent: who can read, who can send commands, scoped to one site on purpose. A read permission never includes control. Safe credential rotation. Provider enforces the rules; Sentinel scores the result. Emits HiveMQ credentials; a Mosquitto dynsec emitter is on the internal list, not promised.

Proof point from the reference architecture: a leaked site credential can publish only its own group and node, never a command. Proven live.

GitOps / Enterprise Governance, \$9,995, pilot, Host

Question it answers: "Who changed what, when, and can we roll it back?"

Configuration as code for the whole estate: signed bundles distributed fleet-wide, review and rollback, an audit trail. First form shipped in 2.8.0. Git adapter and two-key approvals come after 3.0 and are partner-driven. Always say "pilot." Sell it to the customer who already asks for change control, not as a default line.

Host features in these themes (no separate price)

Feature	What it does	Status
Zero-touch enrollment	New site bootstraps identity with a one-time code over a signed TLS channel	2.8.0 slice 1; 2.9.0 CSR/CA, mTLS, HiveMQ credential emit
Capacity intelligence	Host reports its own ceiling from measured evidence and shard count; sparkline; degraded flag	2.8.0, extended 2.9.0
Data lineage (trace)	Six hops from equipment to consumer	2.8.0; wire hops in 3.0.0
Data contracts	Declared shape per stream; strict mode projects bad quality on violation	2.8.0
Tiered route authorization	Read, write, admin tokens on every route, every product; admin fails closed	2.7.0, hardened 2.9.0

SparkValidate, \$2,495 gateway module or \$995 Validate Desk CLI, pilot, any gateway

Question it answers: "We spend weeks hand-typing a validation package that restates what the control system already knows."

SparkValidate reads existing exports (Rockwell .ACD and L5X first, Ignition alarm configuration and journal) and builds one traceable model. It generates 13 documents and protocols with a requirements traceability matrix: URS to FRS to DS to risk assessment to OQ, every ID pointing back to a rung, a tag path or a view. Risk class drives test depth. Alarm rationalization per ISA-18.2 comes from what the alarm journal shows the plant actually doing. Frameworks: GAMP 5, 21 CFR Part 11, IEC 62443. The CLI works offline; the gateway module executes tests against the running system and seals the evidence.

Pilot release, Rockwell systems first. It does not require SparkBridge.

How the three security products fit together

SparkID generates the rules --> Provider enforces them --> Sentinel scores the result and

That sentence sells all three and answers the security architect's question of who does what.

Say this

- "A read permission never includes control."
- "Sentinel observes and changes nothing. The fix is yours to apply."
- "GitOps is a pilot. Here is what ships today and what comes after 3.0."
- "Certificate issuance at enrollment shipped in 2.9.0; reconciliation across the fleet is roadmap."

Never say this

- "FleetOps pushes config to every site." It shows drift; templates and remediation are 3.1.
- "Sentinel is a firewall" or "hardens the network." It assesses.
- "SparkValidate covers Siemens." Rockwell first; say what the site says.

Check

1. A customer asks for change control with approvals. Which product, and what caveat?
2. Which of these are included with Host: SparkID, Sentinel, FleetOps?
3. What does Sentinel change on the network when it runs?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 02 Data consumers](#) ·

04 Drivers and tools

Nine device drivers and four Ignition enhancements. Each is licensed per gateway and works with or without SparkBridge. Thirty minutes. Prices on the [cheat sheet](#).

Learning objectives

- Name the driver for any equipment a prospect mentions and the box it removes.
- Explain the per-gateway rule for standalone items without negative phrasing.

The rule

Drivers and tools are bought on their own, per gateway, at their own price. A site connecting one kind of equipment needs that driver and usually not the others. They are absent from every "included with" list on the site, and the price line says the rest. Do not add "sold separately" or "never bundled"; show the price.

Device drivers, \$695 each per gateway

Driver	Equipment	What it removes	Proof
SparkLogix	Allen-Bradley ControlLogix, CompactLogix over EtherNet/IP	The translator box; direct connect with live tag browse	433 tests, lab smoke on real hardware, field QA in progress. v0.2.0
SparkSNMP	UPSs, switches, PDUs, servers	A separate network monitoring tool for the plant; the network shows up next to the process	One license covers both Ignition lines
IEC 60870-5-104	Utility and substation equipment	The protocol converter and its failure points. Reads only, no control path	Released
DNP3	DNP3 outstations, event data, source timestamps	Another vendor's converter. Reads only	Released
TI 505 / CTI 2500	Older Siemens 505 and CTI 2500 racks	The translator box, its license and its patching	Running on the GreenGuard lab gateway against a CTI 2500
SparkADS	Beckhoff TwinCAT over ADS/AMS, live symbol browse	An OPC UA bridge on the PLC side	Verified against a real TwinCAT runtime on both lines; certificate pinning on by default. First commercial ADS driver for Ignition
SparkS7	Siemens S7-1200/1500 symbolic addressing	Raw DB offsets; plants keep their tag names	Classic S7comm and S7comm-plus (symbolic, optimized DBs, TLS, certificate pinning) verified on S7-1500 in PLCSIM Advanced V7 with a real TIA Portal V20 project through both the 8.1 and 8.3 gateways: browse, reads, subscriptions, writes. Real S7-1500 hardware still pending; say so if asked
Spark61850	Substation IEDs over IEC 61850 MMS, report control blocks	Flattening through a DNP3 gateway; logical-node model intact	Verified byte for byte against libiec61850, buffered reports on a live gateway
SparkBACnet	Building automation over BACnet/IP: browse, read, write,	A separate BMS integration; building equipment in the	Verified against the reference bacnet-stack

	segmented Complex-ACK	same tag tree as the process	
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On SparkS7 and Ignition 8.3: what actually happened

Symbolic S7 addressing worked in the Ignition 8.3 early access builds. It was removed from the 8.3.0 release. IA staff said on the forum (September 2025) that "some licensing issues are preventing us from doing that" and that the issues "came up pretty last minute, and were largely out of Inductive Automation's hands." Symbolic addressing then reappeared as the Siemens Enhanced driver: a separate module, purchased and licensed on its own, not activated by default at install, built on a third-party library. Users complained about the surprise cost for a feature they expected in the platform, about a fixed-delay polling bug that roughly doubled update latency (IA said fixed in 8.3.2 or 8.3.3), about larger tag groups blocking faster ones, a 250-device connection limit, a GLIBC 2.34 requirement from 8.3.3 that rules out Ubuntu 20.04 and Debian 10/11, and S7-1200 G2 support arriving only in 8.3.3. Symbolic access needs S7-1200/1500 and does not cover S7-300/400.

The pitch that stands: "The free 8.3 Siemens driver has no symbolic addressing. SparkS7 gives symbolic S7-1200/1500 addressing on both 8.1 and 8.3 from one \$695 per-gateway license, verified on an S7-1500 in PLCSIM Advanced with a real TIA project, on both gateway lines. Plants keep their tag names instead of raw DB offsets." If asked about real hardware: PLCSIM Advanced runs the same CPU firmware; a site pilot on the customer's own S7-1500 is the offer, not a claim. Do not characterize IA's driver quality; state the timeline and the price line, and let the customer compare. If a prospect already bought Siemens Enhanced, SparkS7 still matters for 8.1 gateways and for one driver across both lines.

Ignition enhancements

Product	Buyer sentence	Status	Price
SparkInflux	"We standardized on InfluxDB and Ignition history has to go there." History storage and history provider, 8.1 and 8.3	Released	\$1,495
SparkValidate	"Weeks of validation paperwork." Module 03 covers it	Pilot	\$2,495 / \$995 CLI
SparkNotify	"The on-call person is in Teams, not in a Vision client." Seven channels, acknowledgment tracked back	Released 0.2.0	\$695
SparkGantt	"Production schedule on a Perspective screen that does not bog down." Virtualized, smooth at 500+ tasks, drag limits enforced on the gateway	Released	\$695

Where a driver enters a SparkBridge deal

A driver sale is often the first purchase. The site gets a direct connection today; SparkBridge Edge comes later when the estate grows. Drivers pass the "benefits multiple customers" test and rarely the architecture ones, so they stay off the SparkBridge roadmap table. Keep them on a separate quote line from the estate.

Say this

- "Licensed per gateway. Works with or without SparkBridge."
- "The translator box comes out, along with its license and its patching."
- "Verified against the reference stack; the report is in the release bundle."

Never say this

- "Included with Edge" or "included with Host" for any driver or tool.
- "Never bundled", "sold separately." Show the price.
- "Writes to the substation." IEC-104 and DNP3 are reads only.

Check

1. A data center has UPSs on SNMP and chillers on BACnet, on one gateway. Quote it.
2. A utility asks whether the DNP3 driver can send controls. Answer.
3. A customer with SparkBridge Host asks whether SparkNotify is included. Answer in the positive form.

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 03 Operate and trust](#) ·

05 Sizing an estate

How to turn a site count into a required set, a server spec and a list price in ten minutes. Forty-five minutes.
Prices on the [cheat sheet](#); full method in docs/REFERENCE-ARCHITECTURE-40.md.

Learning objectives

- Produce the required set and total for any site count.
- Recommend the Host and Provider hardware from the sizing rules.
- Explain HA in one paragraph.

The arithmetic

Required infrastructure first, optional consumers second, so the required number stands on its own. One site hosts the center, so an estate of N sites usually needs N minus 1 Edge licenses plus the Host.

Shape	Lines	Required	Support, optional, per year
40 sites on the customer's broker	39 x Edge \$38,805; 1 x Host \$1,495; broker \$0 from us	\$40,300	\$8,060
40 sites with Provider HA and history	39 x Edge \$38,805; 2 x Provider \$3,990; 2 x Host \$2,990; 1 x SparkVault \$3,995	\$49,780	\$9,956
Complete operating architecture	Required subtotal \$45,785 (39 Edge, 2 Provider, 2 Host) plus SparkVault \$3,995, FleetOps \$4,995, Sentinel \$4,995, GitOps \$9,995 pilot	\$69,765	\$13,953

Site 41 is one more Edge, \$995, an enrollment code, and nothing else. Optional consumers on any shape, none touching the sites: FleetOps \$4,995, Sentinel \$4,995, SparkInject \$2,995 per family, SparkRecord \$7,995, Passage \$1,995. Every figure is list before volume pricing; tell us the site count and we run the numbers.

Generic formula

```
required  = (N - 1) x 995  + hosts x 1,495  + providers x 1,995
hosts      = 1 (single) or 2 (active + warm standby)
providers  = 0 (their broker) or 2 (HA pair); 1 only for a lab
optional  = sum of Host modules chosen
```

support = 0.20 x (required + optional) per year, optional
partner = list x 0.80 (Architecture) or x 0.60 first production estate (Founding); never bo

Sizing rules

Layer	Rule
Edge per site	One license per site gateway. Up to 8 edge nodes. 250,000-tag budget
Wire metrics fleet-wide	Sites x published leaves after metadata folding. 40 x about 7,000 = about 280,000 in the reference estate
Host memory	About 300 bytes per wire metric per consumer. 500,000 wire metrics under 4 GiB. 280k x 4 consumers is about 340 MB of session state plus the managed provider
Host CPU	Projector shards equal core count. Six cores sustained 339,527 updates/s; four logical cores bend at 55,464
Host box	8 cores, 16 GB; 32 GB if SparkVault shares it
Provider	Two HiveMQ CE nodes, 4 cores, 8 GB each. No client-count license
Buffer	50 MB default per edge node is hours to days at typical change rates; 500 MB for cellular and satellite

What happens when things fail

Failure	Behaviour	Measured
WAN drop at a site	Edge buffers to disk; Host marks the site's points bad	2,697 msg/s durable
WAN returns	Reconnect within a 10 s cap, rebirth, ordered replay with the historical flag; SparkVault backfills	400,000 of 400,000; 40,035 upd/s replay on the six-core rig
Host primary crash	Standby promotes after the debounce; edges never rebirth	244 ms across 25 nodes
One Provider node down	Edges and Host walk to the other server	About 0.6 s server walk; MultipleBroker TCK pass
Broker outage entirely	Every site buffers; nothing lost up to buffer size	Store-and-forward suite
A site's credential leaks	It can publish only its own group and node, never a command	Provider ACL, proven live

Worked example: 12 sites, water district

Cellular links, no broker today, wants history that survives outages and a view of every gateway.

Line	Qty	Each	Total
Edge	11	\$995	\$10,945
Provider (single; HA pair when they ask)	1	\$1,995	\$1,995
Host	1	\$1,495	\$1,495
Required			\$14,435
SparkVault	1	\$3,995	\$3,995
FleetOps	1	\$4,995	\$4,995
Total, list			\$23,425
Support, optional, per year			\$4,685

Buffer at 500 MB per site because the links are cellular. SparkLink comes with Edge and answers the data-bill question. Host box: 8 cores, 16 GB, 32 GB since Vault shares it.

Say this

- "Required number first, on its own line. Consumers second."
- "None of these numbers move when the estate gets busier."
- "Not sure of the shape? We size it from your topology in the free architecture review."

Never say this

- A per-tag or per-client figure of any kind.
- "Package" or "suite" totals. Build every quote from lines.
- A discount you have not been given. Volume pricing is Dan's call.

Check

1. 25 sites, they run HiveMQ, want Host HA and SparkVault. Required set and total.
2. Same estate adds Sentinel and the compliance pack. New total and support.
3. What is the Host memory estimate for 25 sites at 13,000 leaves each with 3 consumers?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 04 Drivers and tools](#) ·

06 Licensing

The terms, the mechanics, and what to say when support lapses. Thirty minutes. Prices on the [cheat sheet](#).

Learning objectives

- State the seven terms without notes.
- Walk a customer from trial to licensed without a restart.
- Answer the support-lapse and hardware-move questions.

The terms

Unit	One license per production gateway, per product
Term	Perpetual. Bought once and kept. Not a subscription
Development	Free, no time limit, nothing held back. Development and test gateways are not counted
Support	Optional, 20% of list per year. Covers updates and direct engineering support. Twelve months from purchase, renewal re-issues the key with a new date
Metering	None. Tags, devices and connected clients are never counted
Activation	One signed key file per gateway, applied with no restart
Moving	Follows the role, not the hardware. Move it when you rebuild a server
Volume	Every price is list. Site count goes to Dan for volume pricing
Partners	Architecture Partners 20% off list with deal registration; the first five Founding Partners 40% on their first production estate. Module 09

How the mechanics work

1. **Install.** The customer uploads the signed .modl at Config > Modules. Every Spark product ships signed with the SparkBridge Code Signing certificate on both Ignition lines.
2. **Trial.** Without a key, a paid module runs on Ignition's two-hour module trial. Reset Trial on the gateway restarts it as often as needed. When the clock runs out the data path pauses. SparkLink, SparkCalc and SparkID never pause.
3. **Purchase.** Stripe one-time payment on the site (Buy buttons), or invoice. No subscriptions, ever.
4. **Key.** We issue one signed key file per gateway listing every entitlement bought for it, emailed from admin@greenguard-usa.com. The customer drops it in the Ignition data directory as sparkbridge-

license.key. The module rechecks every 30 seconds, the status page changes from Trial to "Licensed to ...", no restart. CLI tools take `--license <file>`.

5. **Integrity.** A tampered or invalid key fails closed; the product pauses exactly as an expired trial does. Keys are normally issued for any gateway name; a key can be bound to one gateway name on request.
6. **Renewal.** Support renewal re-issues the key with a new support-until date. Nothing changes if it is not renewed.

Questions customers ask

What happens if support lapses?

The software keeps running exactly as before. They stop receiving new versions and support until they renew.

Can we add modules later?

Yes. Each module goes on a Host gateway they already own, on its own license, when the need shows up.

Is development really free?

Yes. Reset Trial as often as needed; a key is bought when a gateway goes into production.

We rebuild the Host on new hardware.

The license follows the role. Move the key file.

Redundant pair: one license or two?

Two Host licenses for active plus warm standby; the pricing page examples show it that way. Two Provider licenses for the HA pair.

We have 8.1 at sites and 8.3 at the center.

Same price, separate downloads per line. One key per gateway either way.

Do you count Ignition Edge edition gateways?

A production gateway is a production gateway. Confirm edition support with engineering before quoting Edge on an Ignition Edge edition install.

Internal: who does what

- Keys are issued by Dan from the signing key on his Mac after payment lands; SEs never issue keys.
- A copy of every issued key is kept in Drive under GreenGuard/SparkBridge/licenses/<customer>.
- Trial-expiry proofs and key-swap proofs were run live on the lab gateway for 2.7.0; the record is in the release bundle if a customer's security team asks.

Say this

- "Per gateway, per product, bought once, kept for good."
- "One key file, thirty seconds, no restart."
- "Nothing stops running if support lapses."

Never say this

- "Subscription", "annual license", "per seat."
- "We can meter it if you prefer." We cannot and will not.
- "Perpetual updates." Updates come with support.

Check

1. A customer edits the key file to add an entitlement. What happens?
2. How long between dropping a key file and the module reporting Licensed?
3. A prospect wants a monthly price. Answer.

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

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07 Discovery

Qualifying questions by vertical, and the routing table from a customer sentence to the product and price. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Run a twenty-minute discovery call that places the estate in a shape and surfaces the first purchase.
- Route any customer sentence to one product line.
- Know when to hand off to the architecture review.

The seven questions, every call

1. How many Ignition gateways, and which are redundant pairs? (shape)
2. How do sites reach the center today: Gateway Network, MQTT, VPN, nothing? (shape)
3. What is the busiest central box doing right now, CPU and memory? (the pain number)
4. What happens to history and alarms when a site's WAN drops? (Edge, Vault)
5. How many consumers of site data exist, and who is asking for the next one? (Host, Passage, Inject)
6. Who builds a new site gateway and how long does it take? (enrollment, FleetOps)
7. Is there a broker already, and who owns it? (Provider or theirs)

Then one vertical question from the table below. Write the answers into HubSpot on the contact before the call ends.

By vertical

Vertical	Ask	Likely first purchase
Multi-site manufacturing	"Which plant is the hub, and what breaks when it goes down?"	Host + Edge, then SparkVault
Water, wastewater, utilities	"What links reach the lift stations, and what does a month of data cost?"	Edge with SparkLink; Vault when the trend gaps come up; IEC-104 or DNP3 driver at substations
Oil and gas	"Who audits your EFM and batch records, and how do you prove a row never changed?"	SparkRecord early access, SparkVault
Pharma, food, validated	"How many weeks does the validation package take, and who types it?"	SparkValidate; SparkRecord for the audit chain
Data centers, facilities	"Is the UPS and the chiller in the same screen as the process?"	SparkSNMP, SparkBACnet on the site gateway
Substations, transmission	"Do the IEDs speak 61850 or are they flattened through a DNP3 gateway?"	Spark61850, IEC-104, DNP3
Beckhoff or Siemens plants	"Are you on raw DB offsets or symbolic names, and is an OPC UA bridge in the path?"	SparkS7, SparkADS
Integrator	"How many estates over ten sites are you scoping this year?"	Partner program, module 09

Routing table

If they say	What changes	What they buy
A remote site on cellular or satellite; link drops; data bill climbs	Report by exception, disk buffer, per-tag spend list	Edge \$995 with SparkLink. Standalone
Forty sites on one central server that is already stretched thin	Sites hold their own data in an outage; the center stops working harder per dashboard	Host \$1,495 plus Edge \$995 per site
A trend runs flat through an outage and then jumps	Every reading keeps its real time; backlog lands where it belongs	SparkVault \$3,995
Analysts reconcile duplicate rows in the warehouse	Exactly once into their own warehouse	SparkInject \$2,995 per family
A historian or SCADA package that will never be replaced	Existing tools connect the way they already do	Passage \$1,995, read only
Three different answers for the same efficiency number	Calculated once, used everywhere	SparkCalc, included with Host
Nobody can say what the network exposes; auditor asking	Scored report with evidence and fix per finding	Sentinel \$4,995; compliance pack \$2,495
One compromised site could write another site's data	Read never includes control; access scoped per site	SparkID, included with Host; Provider \$1,995 if no broker
Forty gateways and nobody knows versions or certificate dates	One view; fix once, apply everywhere	FleetOps \$4,995, worth it from about ten sites
Batch or event records must survive an audit	Append-only, exactly once, SHA-256 chain	SparkRecord \$7,995, early access
Who changed the configuration and can we roll it back	Signed bundles, review, rollback, audit trail	GitOps / Enterprise Governance \$9,995, pilot
An older rack behind a translator box	Box comes out with its license and patching	TI 505 / CTI 2500 driver \$695
A new dashboard or analytics project must not slow the plant	New consumer adds no load at the source	Host \$1,495, or Passage \$1,995 for standards-based tools
Every point live to the whole company, past the tag ceiling	Lossless replayable stream	SparkFlow \$995

When to hand off

Any estate over ten sites, any HA question, any broker the customer already runs, any 8.3-only feature: offer the free architecture review. Engineering sizes it from the topology and returns a written shape with the required number. The review is also the entry condition for the partner program, so an integrator gets it before their first production quote.

Disqualifiers

- No Ignition and no plan to buy it. We sell modules on Ignition.
- One site, no growth, no outage problem, no warehouse. A driver may still fit.
- Wants per-tag or per-client pricing to compare line by line. Explain the model once; do not invent a per-tag number.

Say this

- "Find the row that sounds like your plant. The last column is what fixes it and what it costs."
- "Most plants buy one product for one problem, then add the next when it shows up."

Never say this

- "You need the whole line." Nobody buys everything on day one.
- "We'll figure out pricing later." Every row has a list price today.

Check

1. A plant manager says "our reports are wrong after every storm." Route it.
2. An IT security lead says "I need to know what this MQTT thing is exposing before I approve it." Route it.
3. A customer runs Cirrus Transmission into HiveMQ at 30 sites and wants a warehouse feed. Which question do you ask first, and what is the likely purchase?

► Answers

08 Competitive

What the prospect is comparing us to, what each does well, and where the difference is. Honest by rule: the compare page names what each approach does well and so do we. Forty-five minutes.

Learning objectives

- Position against the three architectures and the four named vendors without disparaging any.
- Answer "we already run Cirrus" in under a minute.

The three architectures

	Traditional distributed Ignition	Module-based MQTT for Ignition	SparkBridge
Shape	Site gateways, Gateway Network, remote tag providers, backends per application family	Sites publish Sparkplug to a broker; one engine gateway subscribes	Edge publishes once; Provider or their broker; Host pool with shared consumers
Does well	Everything at five sites; familiar to every Ignition integrator	Removes polling and most fan-in; site buffering through outages	Everything after the transport: consumer scale-out, provisioning, drift, fleet, policy
Where it hurts	Hub saturation; backend sprawl; history reconciled by hand; provisioning by visit	Second consumer usually means a second engine; credentials typed per site; ACL files by hand; fleet by status page	Early access and pilot labels on Record, Validate, GitOps; no published case study yet; mixed-vendor pairing by specification, bench-proven only for named brokers

Where a module-MQTT cell reads the same as ours, that is Sparkplug doing its job for both. Say it. It buys credibility for the rows that differ.

Named vendors

Cirrus Link (IA's MQTT modules: Transmission, Engine, Distributor, Recorder)

The reference. Transmission is the edge publisher, Engine the primary host, Distributor the broker. Mature, sold through IA, known to every integrator. Our answer is not "they are wrong." It is: same transport, and here is what remains for your team afterwards. Concretely: Engine consumer scale-out (one wildcard subscriber per engine versus our shared subscriptions, four consumers about 20x one), provisioning (typed credentials versus

a one-time enrollment code), fleet (status pages versus FleetOps), policy (ACL files versus SparkID generated rules), history on source timestamps through replay (SparkVault), and exactly-once warehouse delivery (SparkInject; Cirrus Recorder maps to SparkVault in our line). We interoperate: a Host can be primary host for Transmission edges by specification. Say which pairings have run on the bench and which have not.

Cedalo (Mosquitto Pro / Management Center)

Broker vendor. Mosquitto 2.1 added broker-created MQTT 5 topic aliases, SQLite persistence and dynamic security. Cedalo Pro gates HA, UI, identity, audit and integrations, priced per concurrent client (tiers from about EUR 39 to enterprise). Our Provider has no client count, and SparkID generates the policy. SparkBridge has run against Mosquitto 2.1.2 with the persistence plugin on the bench, four tests passing, and fixed a Paho v5 alias edge case doing it. If the customer loves Mosquitto, keep it; we sell Edge and Host on top.

HiveMQ

Enterprise broker; the Provider HA pair in our reference architecture is two HiveMQ CE nodes. Not a competitor to Edge or Host. If the customer runs HiveMQ Enterprise, Provider is optional and SparkID emits HiveMQ credentials. Say "keep it."

Kymera and other driver vendors

Kymera sells InfluxDB and SNMP modules for Ignition among others. SparkInflux and SparkSNMP compete line for line, per gateway, and ship on both Ignition lines. Compete on the proof in the release bundle and on the price line; do not characterize their product.

Ignition-native (do nothing)

Gateway Network, remote providers, redundancy and store-and-forward are all real and all included with Ignition. At five sites, recommend them. The compare page's 5 to 50 table is the honest ramp: the traditional shape is "fine" at 5 and "the overloaded box" at 40.

The one-minute answer to "we already run Cirrus"

"Good. Sparkplug is the right transport and your sites already publish. What we add sits after the broker: one Host namespace with consumers that scale on the shared stream, history and warehouse delivery on source timestamps with exactly-once landing, an enrollment code for site 41, and a fleet view with drift and certificates. Your Transmission edges can stay; we start with Host on your broker. Where we have not run your exact broker on the bench, we say so and run it before you buy."

Where we are honest about limits

- Sparkplug Compatible listing with the Eclipse working group is administrative and in progress; conformance has been run on the TCK harness, "certified" is their stamp.
- Modules are signed with our own code-signing certificate, not an IA-issued one; the gateway asks to trust it once.
- Mixed-vendor edge and host pairing has not run on the bench; the site says so.
- No published production case study yet. The design partner offer exists for that reason. See module 11.
- SparkRecord, SparkValidate and GitOps carry early access or pilot labels.

Say this

- "MQTT is the transport. Operating the estate is the problem."
- "Keep your broker."
- "Every number on our compare page was measured; the reproduction steps are in the reference architecture."

Never say this

- Anything negative about a named vendor's quality or roadmap.
- "Certified" for Sparkplug until the listing exists.
- "Replaces Cirrus." Coexists, and replaces the work after the transport.

Check

1. Prospect: "Cirrus Engine already gives us one namespace." Reply.
2. Prospect: "Mosquitto is free, why pay for Provider?" Reply.
3. Prospect: "Are you Sparkplug certified?" Reply.

► Answers

09 Partner program

Architecture Partners, Founding Partners, the unpublished design partner offer, and the rules for who sends what. Thirty minutes. Public page: mqtt.greenguard-usa.com/sparkbridge/partners.

Learning objectives

- State both partner tiers and the no-stacking rule.
- Know what an SE may draft and what only Dan sends.

Architecture Partner

Who: an integrator with an Ignition practice that has deployed, or is scoping, an estate of ten or more sites and has completed one SparkBridge architecture review with us. Invitation only. Cap of ten.

Benefit	Terms
Development licenses	Unlimited, every module, no time limit (true for everyone; stated so partners know)
NFR licenses	One production-grade key per module for the partner's own demo estate, renewed yearly at no charge
Architecture support	Direct access to the module authors by email and scheduled call for any estate the partner designs
Deal registration	Registered opportunities protected for 90 days; we do not quote the end customer directly on a registered deal
Discount	20% off list on every license the partner resells or deploys
Escalation	Named contact, reply within one business day on production issues
Co-selling	We join technical calls with the partner's customer on request
Training	One remote architecture and deployment session per year, recorded
Roadmap	Quarterly roadmap call; requests logged and answered

Founding Partner

The first five Architecture Partners to take a customer estate into measured production.

Benefit	Terms
NFR keys	Never expire and survive any later change to the program
Priority support	Shared channel with our engineers; production issues first in the queue
Discount	40% off list on the first production estate
Roadmap vote	Founding partners vote on the order of the public roadmap items
Joint case study	We write it, they approve it, both names or anonymized at their request

What we ask of partners

- One completed architecture review before the first production quote.
- Production issues reported with the gateway diagnostics bundle.
- Permission to list the partner by name on the partners page (optional).
- For Founding: a measured writeup from the production estate, per the table on the partners page.

Program limits

- Ten Architecture Partner seats; five Founding.
- Invitation only.
- No discount stacking: 20% or 40%, never both, never with volume pricing on the same line unless Dan says so.

Design partners (end customers), unpublished

A separate offer for end customers, not integrators, kept in docs/gtm/design-partner-offer.md and not on the site. Proposed: 50% off list on the first production estate, 25% off expansion, direct engineering access, the free review plus a written deployment plan and a two-site pilot with us on the call, priority roadmap input, remote training at go-live. The customer provides at least ten sites in production within six months, permission to publish anonymized measurements, up to four reference calls a year, and approval of one case study. Target three to five deployments, two in production, one published case study. Verticals in order: multi-site manufacturing, water and utilities, oil and gas, data centers, distributed logistics. Every figure is "proposed" until Dan ratifies it for a named customer. Never quote it without Dan.

Outreach rules

- Targets: docs/gtm/integrator-targets.csv, 31 firms, mirrored to the "SparkBridge Partner Targets" Google Sheet. Dan approves each row before contact.
- Claude or an SE drafts only; Dan sends. First touch tagged [FAP], follow-up [FAP-2], template in docs/gtm/outreach-first-email.md, signed Dan Fenter, GreenGuard USA.
- Opener that tested best: "What changes when your Ignition deployment reaches 40 sites?"
- HubSpot: contact props outreach_campaign = sparkbridge-fap-2026-09, outreach_segment = vertical; deals in the default Sales Pipeline tagged "SparkBridge FAP".
- Targets for the first wave: 30 contacts, 8 to 12 replies, 5 to 8 technical calls, 3 to 5 reviews, 2 to 3 partners, 1 to 2 pilots.

Say this

- "Architecture Partners take 20% off list with deal registration; the first five Founding Partners take 40% on their first production estate."
- "The review comes first. It is free and it is how we size your customer's estate."

Never say this

- "40% for all integrators." That was corrected on the site on 2026-09-05.
- Any design-partner figure to a customer before Dan ratifies it.
- "I'll send you the agreement." Dan sends.

Check

1. An integrator wants the Founding discount on their third estate. Answer.
2. A partner registers a deal; the end customer calls us directly for a quote. What do we do?
3. What must happen before a partner's first production quote?

► Answers

10 Demo script

A twenty-five minute live demo on the GreenGuard lab gateways that shows the thesis, not the catalog. Rehearse it until the four beats land without notes. Sixty minutes to learn, twenty-five to run.

Learning objectives

- Run the four beats in order: enrollment, link drop, replay, consumer add.
- Know what to click and what number to point at in each beat.
- Recover when a beat fails.

Lab environment

Piece	Where	Notes
Ignition 8.1.54 gateway	localhost:8088 on the GreenGuard Mac	Runs Host, an Edge node, and every Spark module. Shared with engineering; check the calendar and the trial-proof hold notes before touching Reset Trial or reinstalling anything
Ignition 8.3 gateway	gg-wsl (Windows workstation, WSL Ubuntu)	Same modules on the 8.3 line; used for the "both lines" beat and for demos that need a project edit
Broker	Provider on the lab gateway; HiveMQ CE and Mosquitto 2.1 available on gg-wsl	Use Provider for the policy beat, HiveMQ for the "keep your broker" beat
Timescale	gg-wsl, ports 15432 and 15433 over an ssh tunnel	SparkVault and SparkRecord destinations

Ask engineering for the current demo credentials and the read token before the call. Never demo on a customer gateway.

Before the call, ten minutes

1. Open the gateway status page. Every Spark module green and Licensed or in fresh trial.
2. Open Host status: the demo edge node connected, birth timestamp recent, metric count visible.
3. Open a Perspective trend on two demo tags with history through SparkVault.
4. Open the /fleet route in a second tab, the /capacity route in a third.
5. Have the pricing page open in a fourth tab. That is the whole quote when they ask.

Beat 1: a site enrolls, three minutes

Say: "Site 41 is one more Edge license and one code."

1. On Host, Config > SparkBridge Host > Enrollment: issue a one-time code.
2. On the second gateway (8.3 on gg-wsl), Config > SparkBridge Edge > Settings: paste the code, connect.
3. Point at Host: the new node appears with its birth, tags build themselves in the tree. Point at /fleet: the node is registered.

Number to point at: seconds from paste to tags visible. **Caveat to say:** "CSR and mTLS issuance shipped in 2.9.0; fleet-wide reconciliation is roadmap."

Beat 2: the link drops, five minutes

Say: "Every link goes down sometime. Watch what Host does and what Edge does."

1. Cut the edge node's broker connection (engineering provides the switch: a firewall rule on gg-wsl or the Edge Settings disconnect).
2. Point at Host: the site's points go bad, not stale. Point at the trend: it stops, it does not flatline.
3. Point at Edge status: buffer depth climbing, messages per second, disk.

Number to point at: the buffer count growing; "2,697 messages per second durable on the reference rig."

Beat 3: replay, five minutes

Say: "Now the link comes back. Nothing lost, nothing twice, every reading on the time the equipment gave it."

1. Restore the connection. Edge rebirths, replay starts.
2. Point at the trend: the gap backfills with the real curve.
3. Open the SparkVault or SparkInject row view: source_ts earlier than ingest_ts, is_historical true.

Number to point at: replayed count equals buffered count. "Four hundred thousand of four hundred thousand on the bench."

Beat 4: add a consumer without touching the site, five minutes

Say: "Publish once. Add consumers at the center. The site did not change."

1. Show Passage: connect an OPC-UA client (UaExpert on the Mac) to the Host namespace, browse the site's tags. Try a write; it is refused.
2. Show SparkCalc: one calculated efficiency tag derived from two site tags.
3. Show /capacity: Host reporting its own ceiling and headroom with the new consumer attached.

Number to point at: Edge status unchanged; nothing happened at the site.

Close, five minutes

Switch to the pricing tab. Build their required set from lines in front of them (module 05). Offer the architecture review. Stop.

Optional beats by audience

Audience	Beat
Security lead	SparkID: generate rules from intent, show a site credential refused on another site's topic and on any command topic. Sentinel: run the scan, open one finding with its evidence and fix
Data team	SparkInject: the landed Snowflake row with two timestamps four hours apart; SparkRecord: verify CLI walking the chain, "0 breaks"
Operations	FleetOps: versions, drift, certificate dates across the lab gateways
Validated industry	SparkValidate CLI on a sample .ACD: the traceability matrix from rung to OQ
Driver buyer	The CTI 2500 on the lab gateway through the TI 505 driver; SparkS7 against PLCSIM on gg-wsl

When a beat fails

- **Node does not rebirth.** Say what Host is doing (waiting on STATE), give it the 10-second reconnect cap, then show the /fleet row. Do not restart anything on a shared gateway mid-call.
- **Trial paused a module.** Say exactly what it is: "development trial, two hours, a key file lifts it in thirty seconds." Reset Trial only if the hold notes allow it.
- **gg-wsl unreachable.** Run beats 2 to 4 on the single 8.1 gateway with its local edge node; skip beat 1 and describe enrollment with the products page screenshot.
- **Anything else.** The measured numbers on the technology and benchmarks pages are the fallback demo. Never fake a screen.

Never

- Change a customer's gateway during a demo.
- Promise a roadmap date.
- Run a beat you have not rehearsed that week.

Check

1. What is the one thing that must be visibly unchanged at the end of beat 4?
2. Which two timestamps prove replay kept source time?
3. A prospect asks you to point Edge at their test gateway right now. Answer.

► Answers

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11 Objections

Objection cards. Each has the honest answer and the caveat we publish. Read the caveat out loud; it is what makes the answer believable. Thirty minutes.

Learning objectives

- Answer the twelve common objections in under a minute each.
- Never answer with a claim the site does not make.

"What happens to the price when we add tags?"

Nothing. Per gateway, per product. Tags, devices and connected clients are never counted. The 40-site examples on the pricing page do not move when the estate gets busier. *Caveat:* Edge has a 250,000-tag budget per gateway; past that you add an edge node or a gateway, not a fee.

"We already run Cirrus."

Keep the transport. What we add sits after the broker: shared-subscription consumers, history and warehouse on source timestamps, enrollment codes, fleet view, generated policy. Start with Host on your broker. *Caveat:* mixed-vendor edge and host pairing is by specification; we run your exact pairing on the bench before you buy.

"Why pay for Provider when we have HiveMQ / Mosquitto?"

You do not have to. Edge and Host connect through any conformant broker; Provider is for the customer with no broker or who wants generated per-site policy enforced without ACL files, with no client count. *Caveat:* HiveMQ and Mosquitto 2.1 are the brokers that have run on our bench.

"What happens if support lapses?"

The software keeps running exactly as before. You stop receiving new versions and support until you renew. *Caveat:* none; that is the whole answer.

"Early access and pilot: is this production software?"

Edge, Host, Provider, Vault, Inject, Passage, FleetOps, Sentinel, Flow and the drivers are released and live-proven on both Ignition lines. SparkRecord 0.1.0 is early access, SparkValidate and GitOps are pilots, and we label them that way on every page. *Caveat:* buy the labeled ones for the problem they solve today, not for the roadmap.

"Do you have a reference customer?"

Not a published one yet. Every number we print was measured on named hardware with reproduction steps in the reference architecture. The design partner offer exists to produce the first case study, and Founding Partners get a joint one. *Caveat:* never name a customer or imply one. If they need a reference before buying, say so to Dan and route them to the design partner conversation.

"8.1 or 8.3?"

Both. Every product ships signed on both lines and is proven on a live 8.3 gateway before release. Same price. *Caveat:* the 8.3 line is newer; ask engineering about any 8.3-only feature before promising it.

"Are you Sparkplug certified?"

We have run the Eclipse TCK harness and the Sparkplug Compatible listing is in progress. We do not use the word certified until it is granted. *Caveat:* that is the caveat.

"The gateway warned me about the module certificate."

Modules are signed with the SparkBridge code-signing certificate, not one issued by Inductive Automation, so the gateway asks to trust it once. Every release bundle includes the signature and the live proof record. *Caveat:* an IA-issued certificate is on the list; no date.

"Isn't this just MQTT with a markup?"

MQTT is the transport and it is the right one. What you pay for is what remains after the transport: one namespace built from births, consumers that scale on the shared stream, replay on source timestamps into history and warehouse, enrollment, fleet, policy. The compare page lists it row by row and says where the standard alone gets you the same cell. *Caveat:* at five sites, use what Ignition gives you; we say that on the compare page.

"We want a subscription / monthly price."

We sell one-time licenses with optional 20% support. There is no monthly option. *Caveat:* volume pricing by site count is available; that is Dan's call, not a discount an SE offers.

"Who supports this? You are a small company."

Direct engineering support is what the 20% buys: the people who wrote the modules. Every module has a manual, a live proof record and a review record in its release bundle. Development is free so you can prove it on your own estate before you spend anything. *Caveat:* say the team size honestly if asked; do not inflate it.

"Can Passage / IEC-104 / DNP3 write to equipment?"

No. Passage is read only; the utility drivers are reads only. Edge write-back is off by default and allowed only for listed devices and points. Security teams usually count this as a plus. *Caveat:* BACnet and Logix drivers do write; say which is which.

"Your site says Ignition 8.3 dropped symbolic S7, but IA lists a Siemens Enhanced driver."

Both are true. Symbolic addressing was in the 8.3 early access, was removed from 8.3.0 over licensing issues IA said were out of its hands, and came back as the Siemens Enhanced driver, a separately purchased module that is not active by default. SparkS7 gives symbolic S7-1200/1500 addressing on 8.1 and 8.3 from one \$695 per-gateway license, verified on an S7-1500 in PLCSIM Advanced with a real TIA project on both gateway lines (real hardware pilot offered, not claimed). *Caveat:* state the timeline and the price line; do not characterize IA's driver. Module 04 has the detail.

The pattern

Answer, then the caveat, then stop. The caveat is not a weakness; it is the reason the answer is believed.

Check

1. Write the reference-customer answer in your own words, with the caveat.
2. A prospect insists on a per-tag comparison against a competitor. What do you do?
3. Which products carry early access or pilot labels today?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

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Quiz

Forty questions: ten on Ignition, thirty on SparkBridge. Pass mark 80% (32 correct). Closed book except the [cheat sheet](#). Answer key at the bottom.

Ignition (00a, 00b)

1. What is an Ignition gateway, and on what default port is its web configuration served?
2. What is the difference between a tag provider and a remote tag provider?
3. How does Inductive Automation license Ignition: per what unit, and what is not counted?
4. Name two things Ignition 8.3 changed versus 8.1.
5. What does the gateway do when a paid Spark module's two-hour trial expires, and what lifts it?
6. Which Sparkplug message carries metric names and metadata, and which carries only changes?
7. What is the primary host in Sparkplug, and which SparkBridge product plays that role?
8. "Ignition Edge" and "SparkBridge Edge": which is an edition and which is a module?
9. In the hub-with-remote-providers shape, what multiplies hub load and what multiplies site load?
10. What does module-based MQTT leave to custom engineering after the transport? Name three items.

SparkBridge (00 to 11)

11. State the thesis in one sentence.
12. Name the four themes and one product in each.
13. Which two products are the core, and when is Provider optional?
14. What is included with Edge? With Host?
15. List price of Edge, Host and Provider.
16. Edge's tag budget per gateway, and how many edge nodes per gateway.
17. What does Host do to a site's points when that site drops?
18. Five measured numbers that belong to the core, with what each proves.
19. SparkVault versus SparkRecord, in one breath.
20. SparkInject is priced per what, and which family is available first?
21. Can Passage write to equipment?
22. Which of SparkID, Sentinel, FleetOps is included with Host?
23. What does Sentinel change on the network when it runs?
24. Which products carry early access or pilot labels?
25. Price of any protocol driver, and the sentence that describes its licensing.
26. Which two drivers are reads only?
27. Required set and list total for 40 sites on the customer's own broker.
28. Required set and total for 40 sites with Provider HA, Host active plus standby, and SparkVault.
29. What does site 41 cost?
30. Host memory rule of thumb per wire metric per consumer.

31. Support: percentage, term, and what happens if it lapses.
32. How is a license activated, and how long does it take?
33. Does a license follow the hardware or the role?
34. Architecture Partner discount and its condition; Founding Partner discount and its condition.
35. Who sends partner outreach emails?
36. The one-minute answer to "we already run Cirrus," in your words.
37. What do we say when asked for a reference customer?
38. The four beats of the demo, in order.
39. Three phrases the site never uses, and why.
40. A water district with 12 sites on cellular, no broker, wants outage-proof history and a fleet view. Write the quote lines and total.

► [Answer key](#)

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